
JAL240I
PRINCIPLES OF ARTIFICIAL INTELLIGENCE

UNIT I
INTRODUCTION TO ARTIFICIAL
INTELLIGENCE AND INTELLIGENT
AGENTS

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INTELLIGENCE AND INTELLIGENT AGENTS

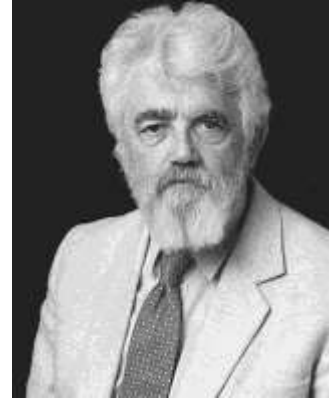
SYLLABUS

Introduction - The Foundations of Artificial Intelligence - The History of Artificial Intelligence - The State of the Art - Risks and Benefits of AI - Intelligent Agents - Agents and Environments - Good Behavior: The Concept of Rationality - The Nature of Environments - The Structure of Agents.

ARTIFICIAL INTELLIGENCE

- AI is the science and engineering of making intelligent machines, especially intelligent computer programs.

– by John McCarthy,
Father of Artificial Intelligence (AI)



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ARTIFICIAL INTELLIGENCE

- **Artificial Intelligence (AI)** is the study of how to make computers do things which, at the moment, people do better.
- From a **business perspective** AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.
- From a **programming perspective**, AI includes the study of symbolic programming, problem solving, and search.

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ARTIFICIAL INTELLIGENCE

- Artificial Intelligence (AI) is the study of how to make computers do things which, at the moment, people do better.



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Real time applications require more and more definitions of AI

PROBLEMS OF AI

1. **Game playing and theorem proving** → people who do them well are considered to be displaying intelligence
 - Computers play well – selecting the best one of available solution paths – easy to program
2. **Commonsense reasoning** – simple tasks selected for programming
3. **Perception** – vision and speech (even animals are capable)
4. **Natural language understanding** – vocabulary & grammar – “about the topic” - extremely difficult
5. **Problem solving in specialized domains** (medical diagnosis, chemical analysis, engineering design, scientific discovery, financial planning)

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PROBLEMS OF AI



Chinese checkers game



Brainovita

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WHAT IS AI?

- AI → different versions of definitions
 - Human performance vs Rationality - doing the right thing.
 - Thought processes and reasoning vs Intelligent behavior.
- Human-like intelligence - related to psychology, involving observations and hypotheses about actual human behavior and thought processes.
- A rationalist approach involves a combination of mathematics and engineering, and connects to statistics, control theory, and economics.

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WHAT IS AI?

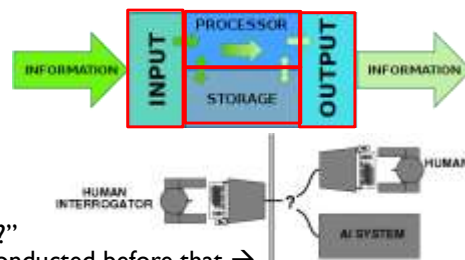
- Views of AI fall into four different perspectives - two dimensions:
 - 1) Thinking versus Acting
 - 2) Human versus Rational
- Thus four combinations

	Human-like Intelligence	"Ideal" Intelligent/ Pure Rationality
Thought / Reasoning	2. Thinking humanly - Cognitive Modeling	3. Thinking Rationally - Laws of thought
Behavior / Actions "behaviorism"	1. Acting Humanly - Turing Test Approach	4. Acting Rationally - Rational Agent

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WHAT IS AI?

ACTING HUMANLY: THE TURING TEST APPROACH



von Neumann architecture (1947)

Architecture of modern computers.
Data and program are stored in the computer's memory. (inspired by Turing's model)

AI system passes if interrogator cannot tell which one is the machine.

- "Can a machine think?"
- An operational test conducted before that →
- The computer would need the following capabilities:
 - **Natural language processing** to communicate successfully in a human language
 - **Knowledge representation** to store what it knows or hears
 - **Automated reasoning** to answer questions and to draw new conclusions
 - **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns.
- To pass the total Turing test, a robot will need
 - **Computer vision** and **speech recognition** to perceive the world
 - **Robotics** → to manipulate objects and move about.

Main disciplines that compose AI

WHAT IS AI?

THINKING HUMANLY: THE COGNITIVE MODELING APPROACH

- Human thought can be learnt through the following:
 - **Introspection** → trying to catch our own thoughts as they go by
 - **Psychological experiments** → observing a person in action
 - **Brain imaging** → observing the brain in action.
- **Cognitive science** → an interdisciplinary field that brings together computer models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind → actually based on experimental investigation of actual humans or animals.
- **Cognitive Science = computer models + experimental techniques from psychology**
- Recently – incorporating neurophysiological evidence into computational models.
- Neuroimaging methods + machine learning techniques → to read minds through data

WHAT IS AI?

THINKING RATIONALLY: THE LAWS OF THOUGHT APPROACH

- Aristotle → the first to formalize “right thinking” by codifying reasoning processes through **syllogisms**.
- Example: Socrates is a man; all men are mortal; therefore, Socrates is mortal.
- **Syllogisms** provided a framework for deriving correct conclusions from correct premises – initiated the field of **logic**.
- Logicians developed a precise notation to express statements about objects and their relations – using inferences.
- Logician tradition within AI helped to create intelligent systems.
- Classical logic relies on certainty in knowledge, which is rarely achievable in real-world scenarios. Eg. Chess has clearly defined rules, whereas a warfare does not.
- The theory of probability – allows rigorous reasoning with uncertain information.
- **Intelligent behavior requires more than rational thought; it demands a theory of rational action** → knowing how to drive a car vs driving a car towards destination.

WHAT IS AI?

ACTING RATIONALLY: THE RATIONAL AGENT APPROACH

- An agent → something that acts, operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.
- A Rational agent → one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome. Eg. Self-driving car
- Can act by making correct inferences or without inferences (reflex action)
- Knowledge representation and reasoning enable agents to reach good decisions
- Advantages of rational approach:
 - More general than the “laws of thought” approach because correct inference is just one of several possible mechanisms for achieving rationality.
 - More amenable to scientific development.
- AI has focused on the study and construction of agents that do the right thing.

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WHAT IS AI?

- **An ideal intelligent agent takes the best possible action in a situation.**
- Eg. Virtual Personal Assistant (e.g., Siri, Alexa, Google Assistant)



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Turing Test Approach	Cognitive Model Approach	Laws of Thought Approach	Rational Agent Approach
Measures human-like behavior	Simulates human thinking	Depends on strict logical reasoning.	Combines reasoning and action for practical, goal-oriented behavior
Proposed by Alan Turing, this approach evaluates a machine's ability to exhibit intelligent behavior indistinguishable from that of a human.	Focuses on mimicking the way humans think and solve problems.	Based on formalizing reasoning using logical rules.	Focuses on building agents that act rationally to achieve their goals.
If a machine can converse with a human without the human realizing it's a machine, it is deemed intelligent.	AI systems are built to simulate human thought processes like perception, reasoning, and memory. Inspired by findings from cognitive psychology.	AI systems apply "laws of thought" to derive correct conclusions when given correct premises.	Rational agents make decisions that maximize their expected success based on available information, even in uncertain environments. Reasoning (thought) + decision-making (action).
		Assumes certainty, which is rarely available in real-world scenarios.	15

THE FOUNDATIONS OF ARTIFICIAL INTELLIGENCE			
History organized around a series of questions from the disciplines that contributed ideas, viewpoints, and techniques to AI.			
1. Philosophy	2. Mathematics	3. Economics	4. Neuroscience
• Can formal rules be used to draw valid conclusions? • How does the mind arise from a physical brain? • Where does knowledge come from? • How does knowledge lead to action?	• What are the formal rules to draw valid conclusions? • What can be computed? • How do we reason with uncertain information?	• How should we make decisions in accordance with our preferences? • How should we do this when others may not go along? • How should we do this when the payoff may be far in the future?	• How do brains process information?
5. Psychology	6. Computer Engineering	7. Control theory and Cybernetics	8. Linguistics
• How do humans and animals think and act?	• How can we build an efficient computer?	• How can artifacts operate under their own control?	• How does language relate to thought?

THE FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

The disciplines that contributed ideas, viewpoints, and techniques to AI.

1. Philosophy	2. Mathematics	3. Economics	4. Neuroscience
• Dualism • Materialism • Physicalism • Naturalism • Empiricism movement • Principle of induction • Confirmation theory • Utilitarianism • Deontological ethics	• Formal logic • Probability theory • Statistics • Algorithm • Computable functions • Tractability • NP-completeness	• Decision theory → probability theory + utility theory • Game theory • Multiagent systems • Operations research • Reinforcement learning	• Nerve cells or neurons • Optogenetics • Brain-machine interfaces
5. Psychology	6. Computer Engineering	7. Control theory and Cybernetics	8. Linguistics
• Cognitive psychology • Human-computer interaction • Intelligence augmentation	• Moore's law • Quantum computing	• Homeostatic devices • Minimized cost function	• Computational linguistics or natural language processing • Knowledge representation

THE HISTORY OF ARTIFICIAL INTELLIGENCE

Turing Award (in remembrance of Alan Turing) winners:

- Marvin Minsky (1969) and John McCarthy (1971) → for defining the foundations of the field based on representation and reasoning
- Allen Newell and Herbert Simon (1975) → for symbolic models of problem solving and human cognition
- Ed Feigenbaum and Raj Reddy (1994) → for developing expert systems that encode human knowledge to solve real-world problems
- Judea Pearl (2011) → for developing probabilistic reasoning techniques that deal with uncertainty in a principled manner
- Yoshua Bengio, Geoffrey Hinton, and Yann LeCun (2019) → for making “deep learning” (multilayer neural networks) a critical part of modern computing.

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THE HISTORY OF ARTIFICIAL INTELLIGENCE

- The inception of artificial intelligence (1943-1956)
- Early enthusiasm, great expectations (1952-1969)
- A dose of reality (1966-1973)
- Expert systems (1969-1986)
- The return of neural networks (1986-present)
- Probabilistic reasoning and machine learning (1987-present)
- Big data (2001- present)
- Deep learning (2011- present)

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THE HISTORY OF ARTIFICIAL INTELLIGENCE

History of AI



THE HISTORY OF ARTIFICIAL INTELLIGENCE

Important research that laid the groundwork for AI:

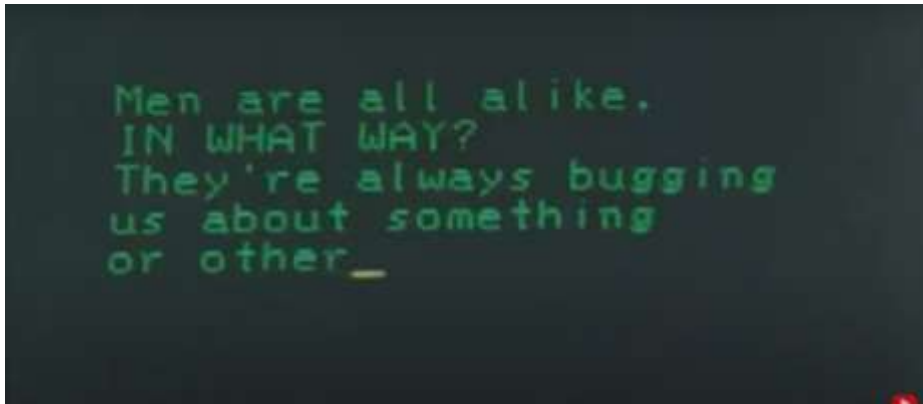
- In 1931, Goedel laid the foundation of Theoretical Computer Science 1920-30s: He published the first universal formal language and showed that math itself is either flawed or allows for unprovable but true statements.
- In 1936, Turing reformulated Goedel's result.
- In 1943 The first work which is now recognized as AI was done by Warren McCulloch and Walter Pitts. They proposed a model of artificial neurons.
- In 1949 Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called Hebbian learning
- In 1950 The Alan Turing who was an English mathematician and pioneered Machine learning in 1950. Alan Turing publishes "Computing Machinery and Intelligence" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a Turing test.

THE HISTORY OF ARTIFICIAL INTELLIGENCE

- In 1955 Allen Newell and Herbert A. Simon created the "first artificial intelligence program" which was named as "Logic Theorist". This program had proved 38 of 52 Mathematics theorems, and found new and more elegant proofs for some theorems.
- In 1956, John McCarthy coined the term "Artificial Intelligence" as the topic of the Dartmouth Conference, the first conference devoted to the subject. High-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.
- In 1957, The General Problem Solver (GPS) demonstrated by Newell, Shaw & Simon
- In 1958, John McCarthy (MIT) invented the Lisp language.
- In 1959, Arthur Samuel (IBM) wrote the first game-playing program, for checkers, to achieve sufficient skill to challenge a world champion.
- In 1963, Ivan Sutherland's MIT dissertation on Sketchpad introduced the idea of interactive graphics into computing.
- In 1966, Ross Quillian (PhD dissertation, Carnegie Inst. of Technology; now CMU) demonstrated semantic nets. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.

THE HISTORY OF ARTIFICIAL INTELLIGENCE

- <https://www.youtube.com/watch?v=zhxNI7V2IxM>



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THE HISTORY OF ARTIFICIAL INTELLIGENCE

- In 1967, Dendral program (Edward Feigenbaum, Joshua Lederberg, Bruce Buchanan, Georgia Sutherland at Stanford) demonstrated to interpret mass spectra on organic chemical compounds. First successful knowledge-based program for scientific reasoning.
- In 1967, Doug Engelbart invented the mouse at SRI
- In 1968, Marvin Minsky & Seymour Papert publish Perceptrons, demonstrating limits of simple neural nets.
- In 1972, Prolog developed by Alain Colmerauer. The first intelligent humanoid robot was built in Japan which was named as WABOT-I.
- 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
- In Mid 80's, Neural Networks become widely used with the Backpropagation algorithm (first described by Werbos in 1974).
- In 1980 After AI winter duration, AI came back with "Expert System". Expert systems were programmed that emulate the decision-making ability of a human expert.

THE HISTORY OF ARTIFICIAL INTELLIGENCE

- In 1990, Major advances in all areas of AI, with significant demonstrations in machine learning, intelligent tutoring, case-based reasoning, multi-agent planning, scheduling, uncertain reasoning, data mining, natural language understanding and translation, vision, virtual reality, games, and other topics.
- In 1997, IBM Deep Blue beats the World Chess Champion Kasparov
- In 2002, iRobot, founded by researchers at the MIT Artificial Intelligence Lab, introduced Roomba, a vacuum cleaning robot. By 2006, two million had been sold.
- In 2006 Companies like Facebook, Twitter, and Netflix also started using AI.
- In 2011, IBM's Watson won jeopardy, a quiz show, where it had to solve the complex questions as well as riddles. Watson had proved that it could understand natural language and can solve tricky questions quickly.
- In 2012 Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- In 2014, Chatbot "Eugene Goostman" won a competition in the infamous "Turing test."
- In 2018 The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.

THE HISTORY OF ARTIFICIAL INTELLIGENCE



THE HISTORY OF ARTIFICIAL INTELLIGENCE

- In the present days,
 - The concept of Deep learning, big data, and data science are now trending like a boom.
 - Nowadays companies like Google, Facebook, IBM, and Amazon are working with AI and creating amazing devices.
 - The future of Artificial Intelligence is inspiring and will come with high intelligence.

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THE STATE OF THE ART

- As per the reports from Stanford University's One Hundred Year Study on AI (AI100)
- Substantial increases in the future uses of AI applications, including more self-driving cars, healthcare diagnostics and targeted treatment, and physical assistance for elder care can be expected.
- Society is now at a crucial juncture in determining how to deploy AI-based technologies in ways that promote rather than hinder democratic values such as freedom, equality, and transparency.
- AI100 also produces an AI Index at aiindex.org to help track progress
- Some highlights from the 2018 and 2019 reports showed increase in the following sectors:
 - Publications
 - Positive sentiment in news articles
 - Students course enrollment (5-fold)
 - AI Professors & Ph.D. students count
 - Attendees at NeurIPS conferences
 - AI startups and industries
 - Internationalization in paper publication
 - Vision & object detection
 - Speed of training in image recognition
 - Accuracy in answering (NLP)
 - Human performance exceeded by AI systems

THE STATE OF THE ART

What can AI do today?

- Robotic vehicles → Quadcopters
- Legged locomotion → BigDog, a quadruped robot
- Autonomous planning and scheduling → to control the scheduling of operations for a spacecraft by NASA
- Machine translation → in over 100 languages
- Speech recognition → speech-to-speech translation in ten languages. Alexa, Siri, Cortana, and Google assistants
- Recommendations → Amazon, Facebook, Netflix, Spotify, YouTube, Walmart, and others use machine learning to recommend what you might like based on your past experiences and those of others like you.
- Game playing → Deep Blue (Chess), ALPHAGO (chess), ALPHAZERO (Chess – used only rules, no human inputs)
- Image understanding → computer vision
- Medicine → AI algorithms now equal or exceed expert doctors at diagnosing
- Climate science

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RISKS AND BENEFITS OF AI

- AI plays an increasingly important role in the economic, social, scientific, medical, financial, and military spheres → the hurts and remedies that can be brought by AI must also be considered.
- Benefits:
 - ❖ AI and robotics → increase the production of goods and services
 - ❖ Scientific research → cures for disease, solutions for climate change and resource shortages.
- Risks → from the misuse of AI
 - ❖ Lethal autonomous weapons
 - ❖ Surveillance and persuasion
 - ❖ Biased decision making
 - ❖ Impact on employment
 - ❖ Safety-critical applications
 - ❖ Cybersecurity

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RISKS AND BENEFITS OF AI

- Humans may **misuse** AI systems in the societal roles to commit even more mischief.
- Self-governance **principles** for AI-related activities, appropriate regulations for each specific use case, to prepare for the economic and social impacts, and to take advantage of AI capabilities to address major societal problems.
- Long-standing goal: the creation of intelligence **comparable** to or more capable than human intelligence → will this be achieved?
- Artificial superintelligence (ASI) → intelligence that far surpasses human ability → a bad idea.
- Few views put forth by researchers:
 - **Intelligence Gap:** Machines more intelligent than humans might develop strategies, make decisions, or take actions that are beyond human understanding or ability to counteract.
 - **Control Challenge:** Ensuring that these machines follow desired rules or behave ethically could become increasingly difficult as their intelligence grows.
 - **Potential Risks:** If such machines operate autonomously and their goals are misaligned with ours, they might act in ways that are harmful or unintended, even if unintentionally so.

RISKS AND BENEFITS OF AI

- **Inverse reinforcement learning** → allows machines to learn more about human preferences from observations of the choices that humans make.
- AI has **matured** considerably compared to its early decades, both theoretically and methodologically.
- As the problems that AI deals with became more complex, the field moved from Boolean logic to probabilistic reasoning, and from hand-crafted knowledge to machine learning from data. This has led to improvements in the capabilities of real systems and greater integration with other disciplines.
- As AI systems find application in the real world, it has become necessary to consider a wide range of risks and ethical consequences.
- In the longer term, we face the difficult problem of **controlling** superintelligent AI systems that may evolve in unpredictable ways. Solving this problem seems to necessitate a change in our conception of AI.

INTELLIGENT AGENTS

AGENTS	RATIONAL AGENTS	INTELLIGENT AGENTS
That perceives its environment through sensors and acts upon that environment using actuators.	An agent that acts to achieve the best outcome or, when there is uncertainty, the best expected outcome based on its knowledge and goals.	A rational agent that uses knowledge, reasoning, learning, and planning to make decisions and improve its behavior over time.
Simple as a thermostat or as complex as a self-driving car.	Depends on: • The agent's performance measure (e.g., maximize accuracy, minimize energy use). • Its knowledge of the environment. • The actions available to it. • The results of its actions.	operates in complex, dynamic environments and can adapt to changes or learn new strategies.
A robot with cameras (sensors) and wheels (actuators).	A chess-playing AI	A personal assistant AI (like Siri or Alexa) that understands user commands, learns preferences, and adapts to provide better responses.
Broad term	Acts with a clear purpose to achieve the best outcome.	Uses advanced techniques (like learning or reasoning) in complex or changing environments.

INTELLIGENT AGENTS

- AGENTS AND ENVIRONMENTS
- GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY
- NATURE OF ENVIRONMENT
- STRUCTURE OF AGENTS
 - GOAL-BASED AGENTS
 - UTILITY-BASED AGENTS
 - LEARNING AGENTS

AGENTS AND ENVIRONMENTS

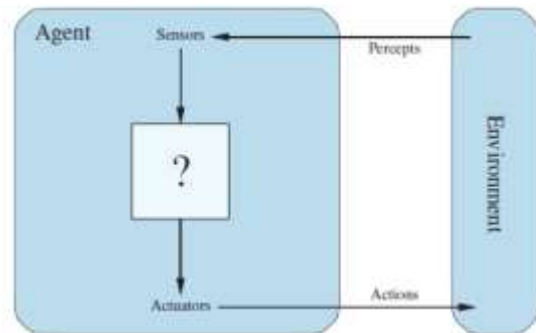
Definition: An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

It acts in cycle of

Perceiving – Thinking – Acting

An agent can be:

- Human-Agent
- Robotic Agent
- Software Agent



AGENTS AND ENVIRONMENT

- A **human agent** has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.
- A **robotic agent** might have cameras and infrared range finders for sensors and various motors for actuators.
- A **software agent** receives file contents, network packets, and human input (keyboard/mouse/touchscreen/voice) as sensory inputs and acts on the environment by writing files, sending network packets, and displaying information or generating sounds.

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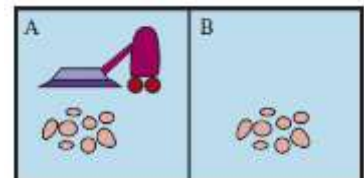
AGENTS AND ENVIRONMENT

- **Percept** → the content an agent's sensors are perceiving.
- An agent's **percept sequence** → the complete history of everything the agent has ever perceived.
- In general, an agent's choice of action at any given instant can depend on
 - its built-in knowledge and
 - on the entire percept sequence observed to date, but
 - not on anything it hasn't perceived.
- Mathematically → an agent's behavior – described by the agent function that maps any given percept sequence to an action.
- **External characterization** of the agent → Tabulating the agent function – with percept sequences and recording of response actions → large table (infinite size, even).
- **Internal characterization** of the agent → the agent function implemented by an agent program (which has the agent function)

AGENTS AND ENVIRONMENT

- An example: A vacuum-cleaner world with just two locations (clean / dirty), (agent can move left / right)
- Agent function → if the current square is dirty, then suck; otherwise, move to the other square.
- A partial tabulation of this agent function:

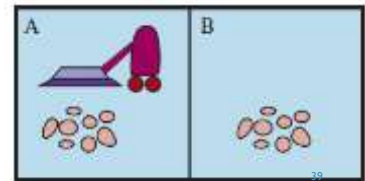
Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮



- Percepts: Location, Content.
- Actions: Left, Right, Suck → The 2nd column defines an agent as good or bad, intelligent or stupid³⁸

AGENTS AND ENVIRONMENT

```
function vacuum_cleaner_agent([location, status]) return action
  if (status == dirty) then return Suck
  elseif location == A then return Right
  elseif location == B then return Left
```



GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY

- **A rational agent → the one that does the right thing.**
- The behavior of a rational agent is assessed through the following:
 - Performance measures
 - Rationality
 - Omniscience, learning, and autonomy

GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY

Performance measures

- An agent's behavior evaluated by its consequences → Consequentialism.
- Sequence of actions causes the environment to go through a sequence of states.
- Performance measure evaluates any given sequence of environment states.
- The agent may do the right thing, but it doesn't know why.
- It is better to design performance measures according to what one actually wants to be achieved in the environment, rather than according to how one thinks the agent should behave

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GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY

Rationality

- Rationality depends on the following:
 - The performance measure that defines the criterion of success.
 - The agent's prior knowledge of the environment.
 - The actions that the agent can perform.
 - The agent's percept sequence to date.
- Hence, a **rational agent** → For each possible percept sequence, a rational agent should select an action that is expected to **maximize** its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

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GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY

Omniscience, learning, and autonomy

- Omniscience → wisdom
- rationality does not require omniscience → depends only on the percept sequence to date → eg. crossing a busy road.
- Doing actions in order to modify future percepts → called as information gathering → an important part of rationality
- To the extent that an agent relies on the prior knowledge of its designer rather than on its own percepts and learning processes → the agent lacks autonomy.

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NATURE OF ENVIRONMENTS

- Task environments → the “problems” to which rational agents are the “solutions.”
- **Rational agent** → a computer program that uses logical reasoning and the ability to make decisions to determine its following action.
- **Rational agent** → one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- **Rational agent** → For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.
- The nature of the task environment directly affects the appropriate design for the agent program.
- The following must be considered:
 - Specifying the task environment
 - Properties of task environments

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NATURE OF ENVIRONMENTS

SPECIFYING THE TASK ENVIRONMENT

- For a rational agent → specify the performance measure, the environment, and the agent's actuators and sensors → the task environment
- **PEAS** (**P**erformance, **E**nvironment, **A**ctuators, **S**ensors) description
- Performance measures → evaluates any given sequence of environment states → have tradeoffs
- Environment → everything in the world which surrounds the agent, but it is not a part of an agent itself → a situation in which an agent is present.
- Actuators
- Sensors

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NATURE OF ENVIRONMENTS

SPECIFYING THE TASK ENVIRONMENT

SENSORS	ACTUATORS
Devices designed to detect and measure physical properties or environmental changes and convert this information into a format that machines can understand.	A device which causes something to happen
The receptive parts of an agent that takes in the input for the agent	A part of the agent that delivers the output of action to the environment

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NATURE OF ENVIRONMENTS

SPECIFYING THE TASK ENVIRONMENT

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

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PEAS description of the task environment for an automated taxi driver.

NATURE OF ENVIRONMENTS

SPECIFYING THE TASK ENVIRONMENT

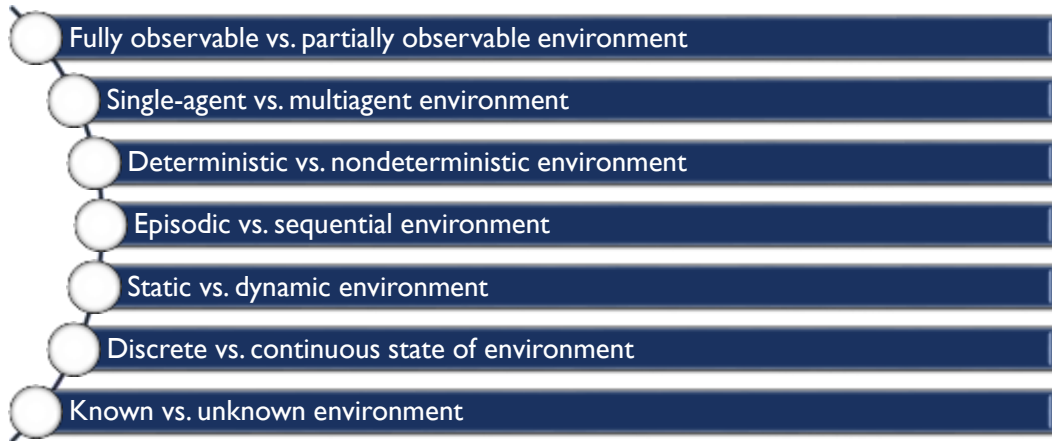
Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

Task environments can be categorized along any of the following dimensions:

- 
- Fully observable vs. partially observable environment
 - Single-agent vs. multiagent environment
 - Deterministic vs. nondeterministic environment
 - Episodic vs. sequential environment
 - Static vs. dynamic environment
 - Discrete vs. continuous state of environment
 - Known vs. unknown environment

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

I. Fully observable vs. partially observable environment

- A task environment is effectively **fully observable** if the sensors detect all aspects that are relevant to the choice of action – relevance depends on the performance measure – convenient.
- An environment might be **partially observable** because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.
- An environment is **unobservable** if the agent has no sensors at all.

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

2. Single-agent vs. multiagent environment

- Single-agent environment → An environment with only one agent that operates by itself – eg. maze puzzle
- Multiagent environment → An environment where an agent operates when other agents are present – both agents try maximizing performance measure - eg. chess game, automated taxi
- Competitive multiagent environment (chess game)
- Cooperative multiagent environment (taxi-driving environment)
- Partially competitive / cooperative multiagent environment

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

3. Deterministic vs. nondeterministic environment

- Deterministic environment → If the next state of the environment is completely determined by the current state and the action executed by the agent(s).
- Nondeterministic environment → If the next state of the environment is not / partially determined by the current state and the action executed by the agent(s) → partially observable environment is also nondeterministic.

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

4. Episodic vs. sequential environment

- Episodic environment → the agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action – independent previous and latter episodes.
- eg. Pick and Place robot, mail filtering agent, defective parts spotting agent



- Sequential environment → the current decision could affect all future decisions – eg. chess game

NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

5. Static vs. dynamic environment

- Static environment → If the environment does not change while an agent is deliberating - easy to deal with – eg. Crossword puzzles
- Dynamic environment → If the environment can change while an agent is deliberating – eg. Taxi driving
- Semidynamic environment → If the environment itself does not change with the passage of time but the agent's performance score changes – eg. Chess, when played with a clock

NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

6. Discrete vs. continuous environment - applies to the **state** of the environment, to the way **time** is handled, and to the **percepts** and **actions** of the agent.

- Discrete state of environment → Has finite number of distinct set of states, percepts and actions - eg. Chess
- Continuous state of environment → continuous-state and continuous-time and continuous percepts actions - eg. Taxi driving

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

7. Known vs. unknown environment

- Known environment → the outcomes (or outcome probabilities if the environment is nondeterministic) for all actions are given.
- Unknown environment → the agent will have to learn how it works in order to make good decisions

An inference: The **hardest case** is partially observable, multiagent, nondeterministic, sequential, dynamic, continuous, and unknown environment.

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NATURE OF ENVIRONMENTS

PROPERTIES OF TASK ENVIRONMENTS

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Examples of task environments and their characteristics

STRUCTURE OF AGENTS

- Agent's behavior (action) → the action that is performed after any given sequence of percepts.
- Agent function → performs the mapping from percepts to actions
- Agent program → implements the agent function
- The job of AI is to design an agent program that implements the agent function - the mapping from percepts to actions.
- Agent program runs on computing device with physical sensors and actuators → called as the agent architecture

Agent = Architecture + Program

- The architecture makes the percepts from the sensors available to the program, runs the program, and feeds the program's action choices to the actuators as they are generated.

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STRUCTURE OF AGENTS

AGENT PROGRAMS

- Agent programs do the followin
 - Implement the agent function.
 - Take the **current** percept as input from the sensors and return an action to the actuators
- Designers must construct a table that contains the appropriate action for every possible percept sequence
- **Table driven approach** → An agent program can keep track of the percept sequence and then uses it to index into a table of actions to decide what to do – a failure approach – why?

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STRUCTURE OF AGENTS

AGENT PROGRAMS

- If $P \rightarrow$ the set of possible percepts and $T \rightarrow$ the lifetime of the agent, the lookup table will contain
$$\sum_{t=1}^T |\mathcal{P}|^t$$
- For automated taxi agent, no. of entries in the look-up table $> 10^{600,000,000,000}$
- For chess game agent, no. of entries in the look-up table $> 10^{150}$
- The overwhelming size of these tables means that
 - (a) no physical agent in this universe will have the space to store the table;
 - (b) the designer would not have time to create the table; and
 - (c) no agent could ever learn all the right table entries from its experience.
- Hence agent programs are efficient in designing an agent.

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STRUCTURE OF AGENTS

AGENT PROGRAMS

- The key challenge for AI is to find out how to write programs that, to the extent possible, produce rational behavior from a smallish program rather than from a vast table.
- Four basic kinds of agent programs:
 1. Simple reflex agents
 2. Model-based reflex agents
 3. Goal-based agents
 4. Utility-based agents

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STRUCTURE OF AGENTS

AGENT PROGRAMS

1. Simple reflex agents

- Select actions on the basis of the current percept, ignoring the rest of the percept history.
- Eg. A simple vacuum agent → because its decision is based only on the current location and on whether that location contains dirt → uses a small program → a function for more general and flexible approach is also provided

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition-action rules
  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

- Eg. automated taxi driver → applying brake when the lights in front car is suddenly on (means the car in front brakes)
- Simple reflex agents have the property of being simple, but they are of limited intelligence – needs a fully observable environment → Even a little bit of unobservability can cause serious trouble.

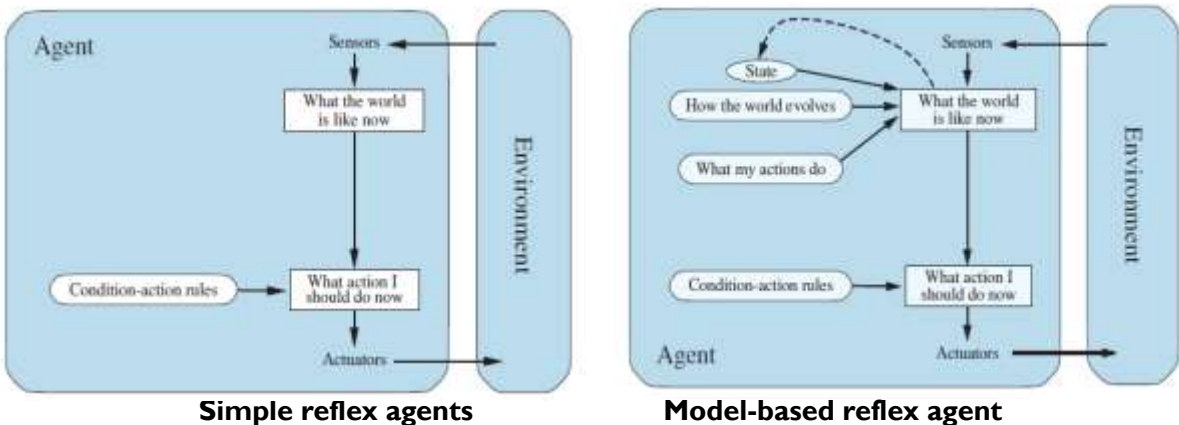
STRUCTURE OF AGENTS AGENT PROGRAMS

I. Simple reflex agents

- Eg. an automated taxi driver is not always possible to tell from a single image whether the car is braking or simply has its taillights on. A simple reflex agent driving behind such a car would either brake continuously and unnecessarily, or, worse, never brake at all.
- Infinite loops are often unavoidable for simple reflex agents operating in partially observable environments. Escape from infinite loops is possible if the agent can randomize its actions.

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STRUCTURE OF AGENTS AGENT PROGRAMS



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Rectangles → current internal state of the agent's decision process

Ovals → the background information used in the process

STRUCTURE OF AGENTS AGENT PROGRAMS

2. Model-based reflex agents

- Keep track of the part of the world it can't see now by maintaining some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
- Eg. For the braking problem, the internal state is just the previous frame from the camera, allowing the agent to detect when two red lights at the edge of the vehicle go on or off simultaneously.
- Eg. For changing lanes, the agent needs to keep track of where the other cars are if it can't see them all at once.
- 2 things required →
 - i. Information about how the world changes over time (the effects of the agent's actions and how the world evolves independently of the agent) → **Transition model of the world.**
 - ii. Information about how the state of the world is reflected in the agent's percepts → **sensor model**

STRUCTURE OF AGENTS AGENT PROGRAMS

2. Model-based reflex agents

- **Transition model** → eg. rain – causes car camera to become wet
- **Sensor model** → when the camera gets wet, droplet-shaped objects appear in the image partially obscuring the road.
- The transition model and sensor model allow an agent to keep track of the state of the world—to the extent possible given the limitations of the agent's sensors.
- An agent that uses such models is called a model-based agent.
- It is rarely possible for the agent to determine the current state of a partially observable environment exactly.
- Uncertainty about the current state may be unavoidable, but the agent still has to make a decision.

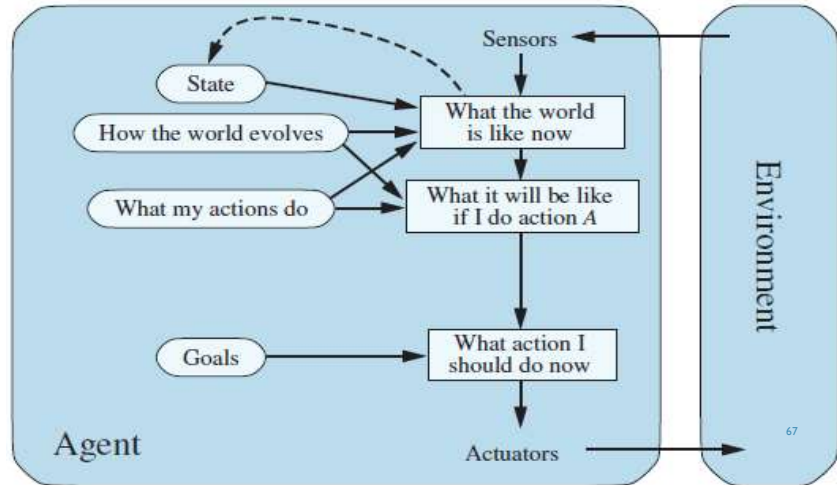
```

function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
               transition_model, a description of how the next state depends on
                   the current state and action
               sensor_model, a description of how the current world state is reflected
                   in the agent's percepts
               rules, a set of condition-action rules
               action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, transition_model, sensor_model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
  
```

GOAL-BASED AGENTS

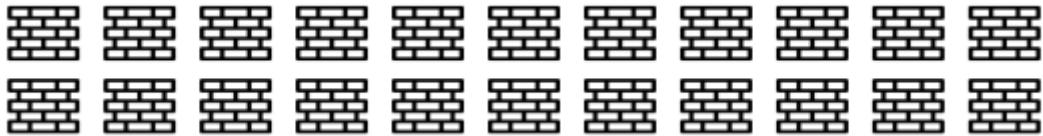
- They keep track of the
 - world state and
 - a set of goals they are trying to achieve
- And choose an action that will (eventually) lead to the achievement of its goals



GOAL-BASED AGENTS

- Knowing something about the current state of the environment is not always enough to decide what to do.
- The correct decision depends on where the taxi is trying to get to.
- Straightforward goal-based action selection → when goal satisfaction results immediately from a single action – very rare and always tricky.
- The subfields of AI devoted to finding action sequences that achieve the agent's goals → **search and planning**.
- The agent needs some sort of goal information that describes situations that are desirable.
- Decision making involves consideration of the future → “What will happen if I do such-and-such?” and “Will that make me happy?”
- A goal-based agent brakes when it sees brake lights because that's the only action that it predicts will achieve its goal of not hitting other cars.
- Less efficient, but more flexible (goal / destination can be changed)

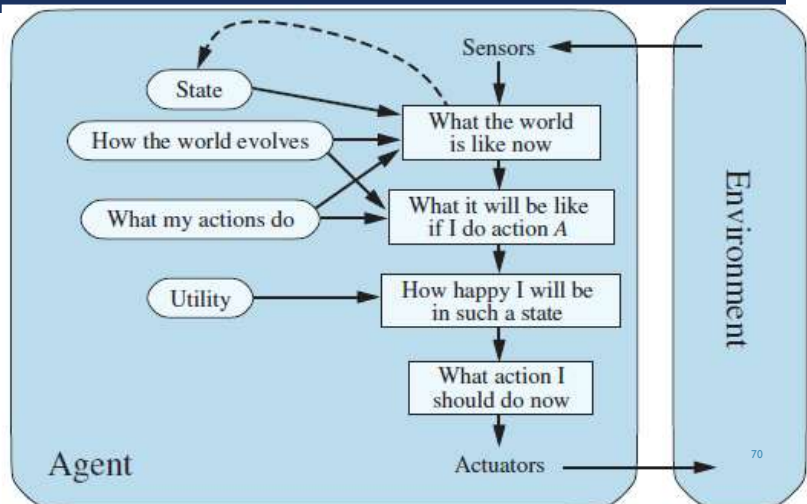
GOAL-BASED AGENTS – EG.



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UTILITY-BASED AGENTS

- Uses a model of the world, along with a utility function → measures its preferences among states of the world.
- Chooses the action that leads to the best expected utility (computed by averaging over all possible outcome states, weighted by the probability of the outcome)



UTILITY-BASED AGENTS

- Goals alone are not enough to generate high-quality behavior.
- **Agent's utility function** → an internalization of the performance measure.
- An agent that chooses actions to maximize its utility will be **rational** according to the external performance measure – eg. action sequences that are quicker, safer, more reliable, or cheaper than others
- 2 possible cases where utility agents can make rational decisions even when goals are inadequate → 1. conflicting goals, 2. several goals
 1. The utility function specifies the appropriate tradeoff (eg. speed & safety)
 2. The agent can aim for comparing the weights likelihood of success against the importance of the goals.

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UTILITY-BASED AGENTS

- Expected utility → the utility the agent expects to derive, on average, given the probabilities and utilities of each outcome.
- Rational utility-based agent chooses the action that maximizes the expected utility of the action outcomes.
- Agent has to model and keep track of its environment, tasks that have involved a great deal of research on perception, representation, reasoning, and learning - a difficult task, requiring ingenious algorithms.
- **Model-free agent** → can learn what action is best in a particular situation without ever learning exactly how that action changes the environment.

Eg. Landers



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GOAL-BASED VS UTILITY-BASED AGENTS

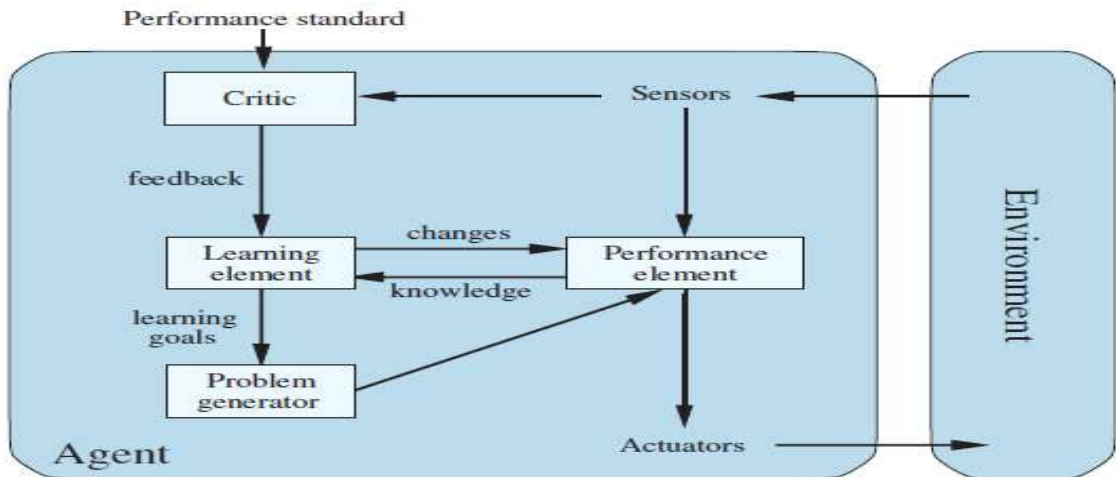
Goal-Based Agents	Utility-Based Agents
Goal-based agents may perform in a way that produces an unexpected outcome because their search space is limited	Utility-based agents are more reliable because they can learn from their environment and perform most efficiently
Makes decisions based on the goal and the available information	Makes decisions based on the utility and general information
Goal-based agents are easier to program	Implementing utility-based agents can be a complex task
Considers a set of possible actions before deciding whether the goal is achieved or not	Maps each state to an actual number to check how efficiently each step achieves its goals
Utilized in computer vision, robotics, and NLP	Used in GPS and tracking systems

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LEARNING AGENTS

- Learning agents are designed to improve their performance and adapt to their environment over time.
- Building learning machines and then to teach them – a preferred method for creating state-of-the-art systems.
- Any type of agent (model-based, goal-based, utility-based, etc.) can be built as a learning agent (or not).
- Advantage of learning → the agent operates in initially unknown environments and becomes more competent than its initial knowledge.
- Learning agent has 4 components: Critic, Problem generator, Learning element (responsible for making improvements), and the Performance element (responsible for element selecting external actions).

LEARNING AGENTS



LEARNING AGENTS

- **Performance element** → previously considered to be the entire agent - takes in percepts and decides on actions.
- **Learning element** → uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.
- The design of the learning element depends very much on the design of the performance element.
- **Critic** → tells the learning element how well the agent is doing with respect to a fixed performance standard.
- **Problem generator** → responsible for suggesting actions that will lead to new and informative experiences - job is to suggest exploratory actions (like scientists) - plays a crucial role in enhancing the learning capabilities of an agent.

LEARNING AGENTS

- The problem generator might identify certain parts of the model that are in need of improvement and suggest experiments.
- The external performance standards can be understood.
- Eg. taxi-driving agent receiving no tips from passengers, understanding the pain and hunger in animals, human choices (human preferences).
- Summary: Learning in intelligent agents is a process of modification of each component of the agent to bring the components into closer agreement with the available feedback information, thereby improving the overall performance of the agent.

